Assignment 4 — Multimedia Technologies (Project Build, Part 2)

> **Notice.** Keep the same team, the same project and the same prototype as in Assignment 3. Start early in Week 8: converting files, measuring sizes, recording and editing a video always take longer than expected, and the video in Part C needs a prototype that already works. Agree inside the team who owns which media assets, which integration task and which part of the video — every member must narrate their own part of the video and be able to reproduce their own work live at the defense.

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1. Purpose

This assignment is the **second practical build stage of your final project**. In Assignment 3 you built the part of your system where the data lives. Here you build the part that people see and hear: the images, icons, sound and video of your prototype — chosen in the right formats, optimised, legally sourced and accessible — and you produce a short multimedia presentation of your working project.

The result becomes part of the system you demonstrate at **Project Stage 2 (End-term, Week 9)** and submit and defend as the **Final Project (Week 10)**. The demo video you make in Part C is also your backup at the End-term: if the network or the cloud service fails during the demonstration, you will be able to show the video instead.

Required reading

- Lecture notes for Week 8 (LMS)
- Parsons, J. J. (2022). *Computer Concepts: Comprehensive*, 21st ed. — **Module 1: Digital Content**
- Nurpeisova, T. B., & Kaidash, I. N. (2017). *Information and Communication Technologies*. Almaty: Bastau

> **Note.** Submission and defense take place in the **Week 8** practice session, as scheduled in the syllabus. Use only free tools if you prefer — every task can be completed with Squoosh (squoosh.app), GIMP, Audacity, HandBrake or ffmpeg, OBS Studio, and your browser's developer tools.

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2. Tasks

Part A — Multimedia formats for your project (20 points)

A1. Format comparison. Compare at least **three formats of each media type** — image (for example JPEG, PNG, WebP, SVG), audio (for example MP3, AAC, WAV) and video (for example MP4 with H.264, WebM, MOV) — in one table:

Format	Media type	Compression (lossy / lossless)	Transparency or animation	Typical file size	Browser and device support	Best use in a web or mobile app
PNG	Image	Lossless	Transparency: yes	Large for photos	All browsers	Logos, icons, screenshots with sharp text
…	…	…	…	…	…	…

Explain in one sentence the difference between a **container** (such as MP4) and a **codec** (such as H.264), since the two are often confused.

A2. Measured experiment on your own material. Take **one real image from your project** — a screenshot of your prototype or a photo it uses — and save it in at least four versions: PNG, JPEG at quality 80, JPEG at quality 40, and WebP. Then take **either a 10-second audio clip or a 10-second video clip** and export it in two formats or two quality settings. For every version record:

File	Format and setting	Resolution or bitrate	File size (KB)	Visible quality (sharp / acceptable / poor)
menu_photo.png	PNG	1280 × 853	…	Sharp
…	…	…	…	…

Show the image versions **side by side at 100% zoom** in one screenshot, so the quality difference can actually be seen. Below the table, write **100–150 words** stating which format and setting your project will use for which kind of content, and why — refer to your own numbers.

Part B — Multimedia in the prototype (30 points)

B1. Media asset inventory. List every media asset your prototype uses or will use — at least **six**, for example a logo, an

icon set, product or menu photos, illustrations, a notification sound, an onboarding animation or a short help video:

#	Asset	Where it is used	Format	Resolution or duration	File size	Source	Licence
1	App logo	Header, browser tab	SVG	Vector	4 KB	Own work	Own
2	...	...	...	...	...	...	...

Every asset must be legally usable. Accept only: your own work; public-domain or CC0 material; Creative Commons material with the required attribution; or royalty-free libraries whose licence you have read (for example Unsplash, Pexels, Pixabay, the YouTube Audio Library). Images found through a search engine are **not** free to use. AI-generated images are allowed only if you state the tool used and the instructor has permitted AI use.

B2. Integration and optimisation. Integrate media into the **running prototype from Assignment 3**. The prototype must contain at least:

- **optimised raster images** in the format you chose in Part A, at a size suited to the screen (no 4000-pixel photo shown at 300 pixels), displayed correctly at 360 px width;
- a **vector icon set or logo** (SVG or an icon font);
- **one time-based media element**: a notification sound, a short embedded video or animation (for example an onboarding or how-to clip), or an animated element (GIF, CSS animation or Lottie).

Then **measure the effect**. In your browser's developer tools, open the Network tab, disable the cache, set throttling to "Fast 4G" and reload your main screen **before and after** optimising the images. Attach both screenshots and fill in:

Measurement	Before optimisation	After optimisation	Change
Total transferred size	...	...	...
Number of requests	...	...	...
Load time (Finish or Load)	...	...	...

State whether your main screen now meets the **performance requirement** you wrote in Project Stage 1, Part B.

B3. Accessibility of media. Show, with a code snippet or a screenshot of each:

- **alternative text** for every meaningful image (and empty `alt=""` for purely decorative ones);
- **captions or a transcript** for any video or audio with speech;
- that **no information is given by sound alone** — every sound has a visual equivalent;
- that **nothing plays sound automatically**;
- sufficient **contrast for any text placed over an image** (at least 4.5:1).

Part C — Multimedia presentation of your project (30 points)

Produce a **2–3 minute demo video** of your project. This is the "simple multimedia presentation" of Week 8, and it will serve as your backup at the End-term.

C1. Storyboard. Before recording, plan the video as a table of **6–10 shots**:

Shot	Duration	What is on screen	Narration (who speaks and what)	On-screen text or graphic
1	0:00–0:15	Title card with project name and team	Member 1: the problem in one sentence	Survey figure from Stage 1
2	...	...	...	...

C2. The video. The video must:

- show the **real, running prototype** — a screen recording of a user completing the main task, not only slides;
- include at least one **diagram or graphic**, for example your architecture diagram;
- have **voice narration by every team member**, each explaining the part they built;
- carry **captions or subtitles** for all speech;
- be **legible**: text on screen readable on a laptop at normal size, no tiny screen recordings;
- use only media you are allowed to use (Part B1 applies to music and images in the video too);
- show **no real personal data** — use test accounts, and blur anything private.

Export it as **MP4 (H.264 video, AAC audio)**, 1280 × 720 or 1920 × 1080, 30 frames per second. Upload it to the LMS; if it exceeds the upload limit, upload it to Google Drive, OneDrive or YouTube (unlisted) and give the instructor access.

C3. Production note. In **80–120 words**, state the tools you used to record and edit, and your export settings — container, codec, resolution, frame rate, bitrate and final file size — and explain why you chose them rather than a higher or lower setting.

Part D — Integration with the project (10 points)

D1. Updated prototype. Submit the current source code or project link with a short note on how to run it. The prototype must include all the media from Part B and still perform the read and write operations from Assignment 3.

D2. Change log. A short table of what changed since Assignment 3 — features added, requirements revised, design decisions changed after testing — and the reason for each change. Three to six rows is enough.

D3. Contribution table. One row per team member stating which media assets, which integration tasks and which part of the video that member personally produced.

Part E — Defense (10 points)

Present your work for **4–6 minutes** in the Week 8 practice session, including **playing part of your video**, and answer **2–3 individual questions**. Every team member speaks about the part they produced.

> **Note.** The instructor may ask you to **demonstrate the work live**: open the developer tools and show the page weight of your main screen, convert an image to another format on request and explain the change in file size, show the alternative text of an image in your code, or open the original project file of your video. Bring a laptop with the prototype, the media files and the tools you used.

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3. Deliverables checklist

☐ Report with Parts A–D, exported as a single PDF named `Group_Team_ICT_Assignment4.pdf`

☐ Format comparison table, and the measured experiment with the side-by-side image screenshot

☐ Media asset inventory with a source and licence for every asset

☐ Before-and-after network screenshots with the measurement table

☐ Evidence of alternative text, captions and the other accessibility checks

☐ Storyboard, production note, and the demo video (MP4 file or a link with access for the instructor)

☐ Prototype source code or project link, with instructions to run it

☐ Change log and contribution table naming what each member produced

☐ Reference list, including attribution for every Creative Commons asset (any consistent style — APA, IEEE or the style used in your textbook)

☐ Uploaded to the LMS before the deadline; one submission per team, with all members named

☐ Every member ready to reproduce their own part live

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4. Grading rubric (100 points)

Criterion	Excellent (90–100%)	Good (70–89%)	Satisfactory (50–69%)	Fail (0–49%)	Max
A. Multimedia formats	Accurate comparison of nine or more formats with container and codec distinguished; measured experiment on own material with sizes and a visible quality comparison; conclusion links each format to a purpose using own numbers	Comparison mostly accurate; experiment present but limited, or quality not compared visually	Generic comparison copied from sources; no measurement on own material	Comparison missing, or lossy and lossless confused	20
B. Multimedia in the prototype	Complete inventory with valid licences; three media types integrated in the running prototype; page weight measured before and after with a real improvement; accessibility checks shown throughout	Media integrated with minor gaps; measurement present but incomplete; some alternative text or licences missing	Media added without format reasoning; no measurement; sources or licences unknown	No media integrated, or copyrighted material used without permission	30
C. Multimedia presentation	Planned storyboard; 2–3 minute video showing the real prototype working; clear narration by every member; captions; legible screens; export settings explained and appropriate	Complete video with minor issues — uneven audio, pacing, partial captions, or one member not narrating	Slides without the running prototype, or no narration or captions, or screens too small to read	No video, or the video is not about the project	30
D. Integration with the project	Prototype runs with all media and the Assignment 3 operations; change log explains decisions; contributions clear and balanced	Prototype runs with small gaps; change log or contributions partly vague	Prototype incomplete or media not integrated; contributions not identifiable	No working prototype	10
E. Defense	Confident live demonstration; answers accurately, including a new task asked on the spot	Demonstrates the work; minor hesitation on details	Explains only part of the work; cannot reproduce anything live	Cannot run or explain own submitted work	10
				Total	**100**

Formatting, citation quality and readability are assessed inside criteria A–D. Work that cannot be defended orally is graded as the defense score alone, regardless of the written quality.

In team projects the score is awarded individually where contributions differ: a member who cannot explain or reproduce the part attributed to them in the contribution table receives a reduced mark for the corresponding criteria.

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5. Sample defense questions

1. Why is JPEG a poor choice for a logo with sharp edges and a transparent background? What would you use instead?
2. Explain lossy and lossless compression using the file sizes you measured.
3. Your main screen became smaller after optimisation. Exactly where did the savings come from?
4. What is the difference between a container and a codec? Which ones does your video use?
5. Why did you export your video at this resolution and bitrate rather than a higher one?
6. Read me the alternative text of this image. Would a screen-reader user understand the screen from it?
7. Where did this photo come from, and what does its licence allow you to do with it?
8. A student uses your app in a lecture with the sound off. What do they miss, and how did you prevent that?